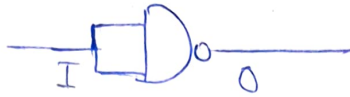


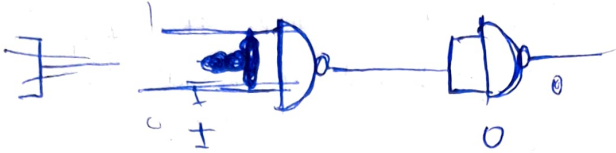
M1 :-

(a) NOT GATE

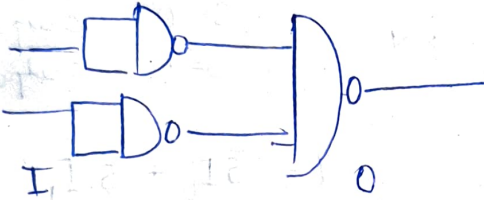


I	O
0	1
1	0

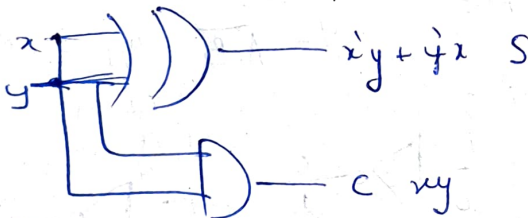
(b) AND GATE



(c) OR GATE



M2 :-



(a) Truth Table

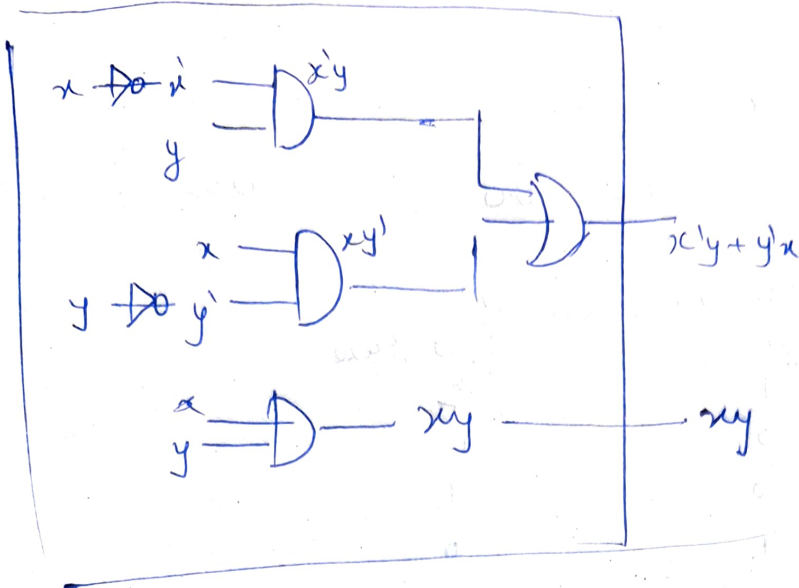
x	y	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

(b) Boolean exp.

$$S = x'y + y'x$$

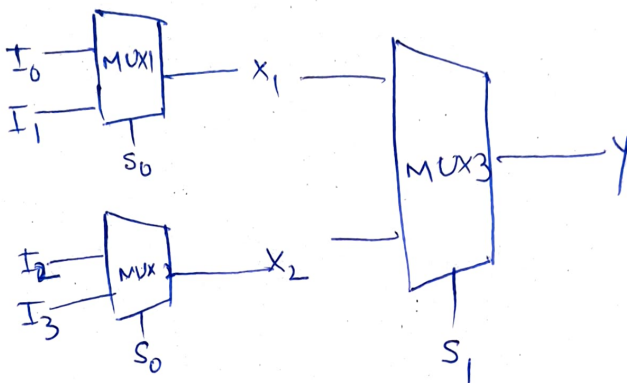
$$C = xy$$

(c) circuit



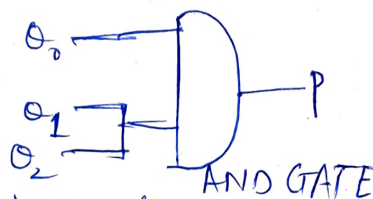
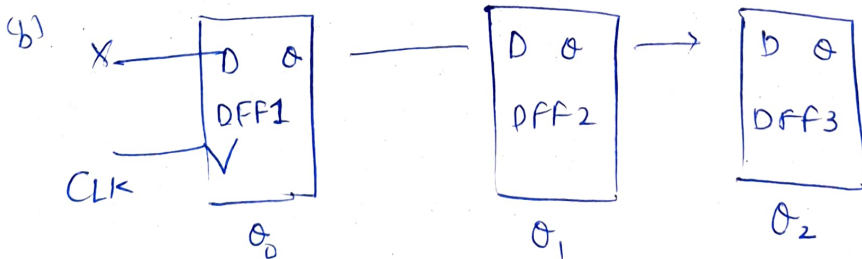
M3:-

Soln:-



M4:-

(a) 3 D-FlipFlops as to know 3 1's it need to store atleast 3 inputs



$Q_2 \rightarrow$ eldest bit $Q_1 \rightarrow$ middle bit $Q_0 \rightarrow$ recent bit

For $P = 1$ $Q_0 = Q_1 = Q_2 = 1$

$P = Q_0 \cdot Q_1 \cdot Q_2$

M5:-

ⓐ

(b) Boolean Expression

$$D0 = \overline{00}$$

where $D = 0+$

(a) State Transition Table

C.S

θ_1

0

0

1

1

Next state

θ_2

0

1

0

1



From the 2-to-1 MUX, the output is 01, which is connected to the 4-to-1 MUX. The output is 11, which is connected to the 2-to-1 MUX. The output is 00, which is connected to the 4-to-1 MUX. The output is 10, which is connected to the 2-to-1 MUX.

